Ashwin Conrad

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Mechanical engineering student focused on vehicle systems, manufacturing, testing, and build-ready design. Experience in Formula EV low-voltage integration, industrial control-panel QA, engineering analysis, and Python automation.

EDUCATION

University of Alberta | Faculty of Engineering, BSc in Mechanical Engineering - Edmonton, AB | **2023 - Present**

West Island College | High School - Calgary | **2020-2023**

- Played varsity Basketball & Rugby; concurrent with Club Basketball and volunteer coaching Youth Basketball.
- Active Leader in the Young Men & Diversity Equity and Inclusion Clubs supporting an inclusive and multicultural community.

CORE CAPABILITIES

Vehicle Systems Integration	**Control Panel Assembly & QA**	**Manufacturing & Test**
CAD & Engineering Drawings	**Python & Engineering Data**	**Cross-functional Coordination**

EXPERIENCE

Integrity & Reliability Engineering Student **2025 – 2026**

AltaGas - Calgary, AB

- Worked with the Asset Integrity, and Reliability teams to provide technical support to facilities in maintaining fixed and rotating equipment.
- Developed a Power Bi with a Microsoft Fabric data pipeline to visualize and monitor vendor corrosion data for over 200 coupons.
- Leveraged python to screen and model operating data gathered by the Pi Historian to provide context for engineering decisions.
- Coordinated the tracking of over 10,000 assets flagged as safety critical elements based on criticality mapped from facility shutdown keys.
- Developed a vessel flare volume blowdown calculator to provide operators with an estimate to compare metered volumes against

Control Panel Technician **Summer 2023 & 2024**

Spartan Controls - Calgary, AB

- Gained hands-on experience in industrial controls working in two departments: Electronics and Remote Transmission Units, and gained exposure to MCCs, VFDs, and the industrial manufacturing process consistently ranked top 50 best managed private companies.
- Assembled, wired, loaded, and performed quality reviews on 30+ industrial control panels from drawings to ready for delivery.
- Loaded software updates on 250+ PLC in under half the allocated time; this was completed while shadowing the Test-Bench team where I supported function checks, and hydrostatic pressure tests on metering panels.
- Supported a safe working environment while completing repeatable assemblies and quality reviews in a fast-paced environment.
- Completed projects in individual, partner-based, and group setups, consistently exceeding expectations and identifying ways to make the process more efficient, ensuring completion under strict deadlines.

LEADERSHIP

Low Voltage Electronics Lead **Winter 2025 - Present**

Formula SAE EV - University of Alberta

- Lead the UA-26 Low-Voltage subsystem remotely by focusing on new-member onboarding and transparent documentation.
- Coordinated design goals, integration, and manufacturing with other subsystems to ensure a shared knowledge base.

Engineering Student Engagement Leader **2024-2025**

University of Alberta - Edmonton, AB

- As an Engineering Student Engagement Leader, I supported the faculty in hosting open houses, first year events, and community events.

COMMUNITY INVOLVEMENT

Youth Basketball Coach **2023-2026**

CYDC - Calgary, AB

- Working with groups of 20-40 kids, ranging in age from 8-17 years old, during weekly practices over 4 month seasons
- Focused player development on teamwork, communication, fundamental basketball skills, agility, and general physical fitness.

RECOGNITION

Undergraduate Student Leadership Award **2024-2025**

Leadership in Engineering Scholarship **2023-2024**

APEGA Science Olympics Medalist: Silver **2021-2023**

Ashwin Conrad | PROJECT PORTFOLIO

ashwin-conrad.github.io

PROJECT PORTFOLIO

Industry Projects

Brazed Aluminum Heat-Exchanger Lifecycle Analysis **2025 – 2026**

AltaGas - Python, PI Historian Excel add-in

- Analyzed 2+ years of operating data for 12+ brazed aluminum heat exchangers.
- Screened lifecycle evidence against OEM limits and summarized inspection and operating-awareness recommendations.

Inspection Scope Generation Tool **2025 – 2026**

AltaGas - Python, openpyxl, python-docx

- Developed a Python tool for equipment-specific inspection-scope generation.
- Generated 500+ standardized scopes from database exports using openpyxl and python-docx.

Course Projects

ENG 160 Engineering Design Project **First Year Project**

University of Alberta - Mechanical design process, concept development, technical documentation

- Lead a five-man team to develop an educational device to model the scientific principles behind a pendulum.
- The device was designed to be assembled by elementary age children and is stored flat to minimize storage space.
- Designed and manufactured a custom needle bearing from wooden dowel, PVC pipe, and stainless-steel pins to minimize friction.
- Our efforts were acknowledged as the project was nominated for top first year engineering project.

MEC 260 Mechanical Design Project **Second Year Project**

University of Alberta - Mechanical prototyping, packaging, mechanism iteration

- Lead a team of six mechanical engineering students through development of a prototype fruit picking and delivery vehicle.
- Iterated physical packaging and mechanism performance through CAD and a linkage motion generator before any physical model.
- Developed team and leadership skills as challenges were identified
- Reinforced the importance of testing and validation in the engineering process as issues were identified to late in the design cycle.

Extracurricular Projects

2026 Low-Voltage Systems and Harness Design, Manufacturing, and Development **2026**

Formula EV Racing - Low-voltage electrical systems

- Designed and developed the 2026 low-voltage system and vehicle harness architecture.
- Supported harness manufacturing, integration, serviceability, and cross-team development.

Personal Projects

Leather Tool Carrier **2024**

Personal Project - Shop workflow tooling

- Designed a leather tool carrier for common hand tools used during panel assembly.
- Built a durable shop aid that kept frequently used tools organized and accessible improving the efficiency of the assembly process.

Composite Skateboard Decks **Earlier build series**

Personal Project - Lamination, fiberglass layups, embedded aluminum

- Built composite skateboard decks through hands-on lamination and fiberglass layup.
- Experimented with embedded aluminum and finish work across an early build series.

TECHNICAL SKILLS

CAD & DESIGN	SolidWorks, AutoCAD & 2D laser cutting, Engineering drawings, 3D modelling and printing
VEHICLE & ELECTRICAL	Low-voltage systems, Power distribution, Sensor integration, Electrical schematics, Panel assembly, Pre-delivery QA
DATA & AUTOMATION	Python, Excel data analysis, Power BI, Microsoft Fabric, Power Query, openpyxl, python-docx
ENGINEERING SOFTWARE	Aspen HYSYS, Maximo, JDE, Pinnacle IDMS, AVEVA Pi Systems
FABRICATION & QA	Engineering drawings, Panel assembly

Expanded case studies and technical work: **ashwin-conrad.github.io**